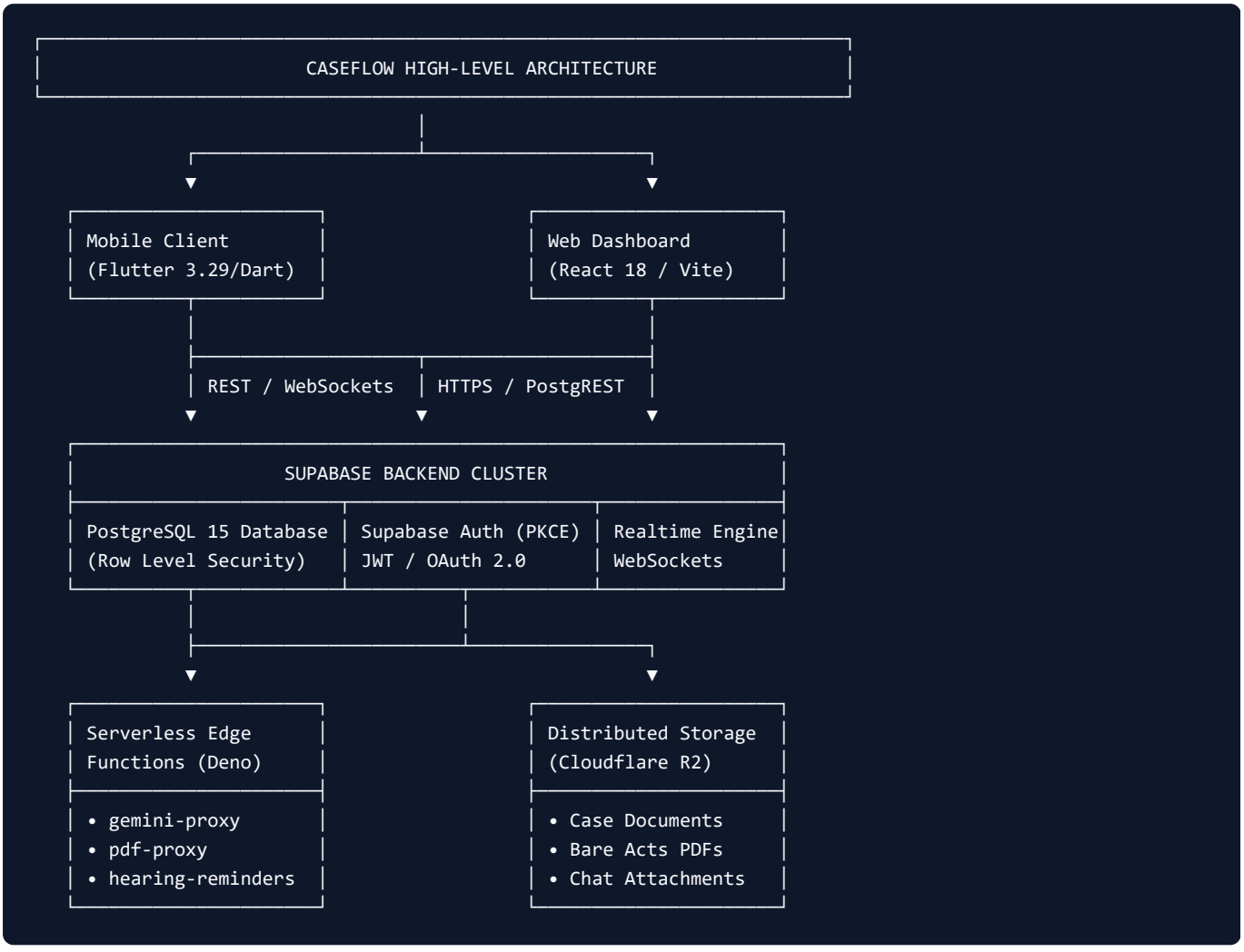
CaseFlow — Technical Requirement Document (TRD)

Version 2.5 · System Architecture & Engineering Specification

Confidential — For Core Development & Infrastructure Engineers

1. System Architecture Overview

CaseFlow employs a modern **Distributed Client-Server & Edge Architecture**. The client layer is a reactive Flutter cross-platform mobile application utilizing Riverpod state management and SQLite offline-first storage. The backend layer relies on Supabase (PostgreSQL, Realtime, Auth) combined with Deno-based Edge Functions and Cloudflare R2 object storage.



2. Technology Stack & Key Dependencies

2.1 Mobile Application Layer (Flutter Engine)

- **Framework:** Flutter 3.29.x / Dart 3.7.x
- **State Management:** Flutter Riverpod (`StateNotifierProvider` , `FutureProvider`)
- **Routing:** `go_router` 14.x with declarative guards
- **Responsive Layout:** `flutter_screenutil` (Design baseline: `390×844dp` , clamping text scale: `0.8x–1.15x`)
- **Local Database:** `sqflite` 2.4.x (SQLite engine)
- **Push Notifications:** Firebase Cloud Messaging (`firebase_messaging` 15.x) & `flutter_local_notifications`
- **Telemetry & Logging:** `firebase_crashlytics` & `firebase_analytics`

2.2 Backend & Cloud Infrastructure

- **Database Platform:** Supabase Managed PostgreSQL 15.x
- **Authentication:** Supabase Auth (PKCE flow with Refresh Tokens)
- **Realtime Engine:** Supabase Realtime WebSockets (`supabase_flutter` 2.x)
- **Serverless Runtime:** Deno 1.x (Supabase Edge Functions)
- **Object Storage:** Cloudflare R2 (S3-compatible API, 0 Egress cost model) + Supabase Storage

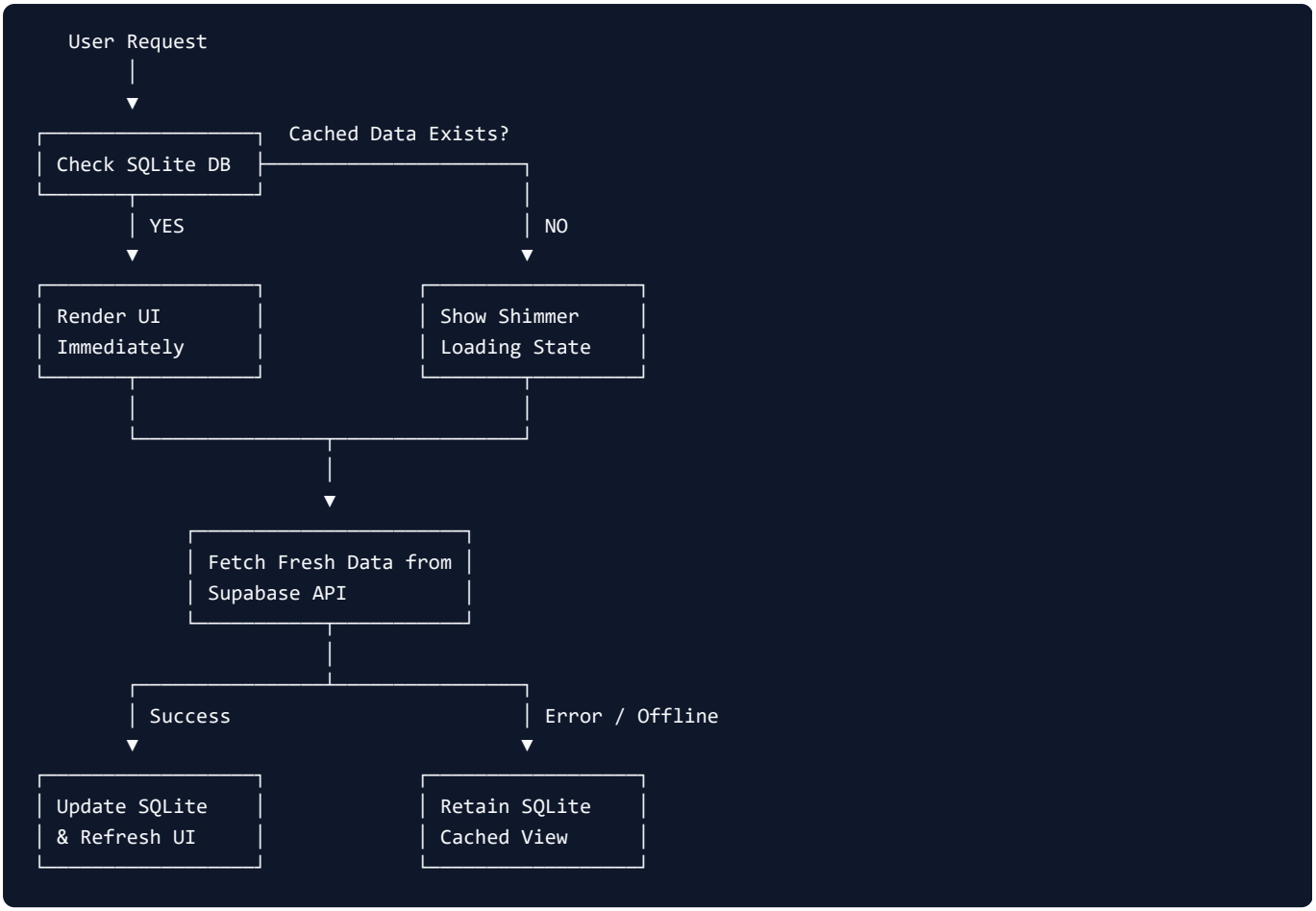
3. Client Architecture & Data Flow

3.1 Repository Pattern & Layering



```
│              SQLite Engine (sqflite)                    │
└─────────────────────────────────────────────────────────┘
```

3.2 Offline-First Cache Design ("Cache-Then-Network")



The `OfflineCacheService` handles local database operations with table definitions for `cached_cases`, `cached_hearings`, `cached_clients`, `cached_stats`, and generic key-value pairs with configurable TTL.

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4. Backend Service Specifications

4.1 Serverless Edge Functions

A. `gemini-proxy`

- **Purpose:** Offloads Gemini 2.0 Flash AI API key handling to the server. Accepts legal prompts, injects system prompts for Indian Legal context, and streams text back to client.
- **Security:** Validates Supabase JWT; API key stored in Deno Secret Store (`GEMINI_API_KEY`).

B. `pdf-proxy`

- **Purpose:** Handles PDF compression, conversion (Word to PDF), merge, and OCR processing using iLovePDF backend APIs.
- **Security:** Protected by secret key validation (`ILOVEPDF_PUBLIC_KEY`, `ILOVEPDF_SECRET_KEY`).

C. `send-hearing-reminders`

- **Purpose:** Cron trigger executing every morning at 08:00 AM IST. Queries hearings scheduled for today, builds personalized FCM v1 HTTP API notification payloads, and dispatches push alerts to registered advocates.

D. `r2-storage`

- **Purpose:** Generates signed S3 upload/download URLs for Cloudflare R2 buckets, enabling large file transfers bypassing application server bandwidth limits.

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5. Security & Cryptographic Integrity

5.1 Row Level Security (RLS) Paradigm

All database access is guarded by Postgres RLS policies based on `auth.uid()`.

```
-- Example RLS Policy for Cases Table
CREATE POLICY "Users can access own or firm cases" ON cases
  FOR ALL USING (
    created_by = auth.uid()
    OR assigned_to = auth.uid()
    OR firm_id = (SELECT firm_id FROM profiles WHERE id = auth.uid())
  );
```

5.2 Evidence Chain-of-Custody Hashing

When evidence photos are captured via the Evidence Lab or Commissioner module: 1. EXIF metadata is recorded (GPS coordinates, altitude, timestamp). 2. The image byte stream is hashed locally using **SHA-256**. 3. The resulting 64-character hash is saved alongside the image metadata in Supabase DB before storage upload, guaranteeing court admissibility.

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6. Performance Benchmarks & Targets

Metric	Target Baseline	Measured (Prod Build)
App Cold Start	< 2.0s	1.45s
Dashboard Load (Parallel)	< 800ms	320ms (Cache), 610ms (Network)
Offline Fallback Latency	< 50ms	18ms
Frame Render Rate	60 FPS	59.4 FPS (Average)
APK Binary Size	< 45 MB	38.2 MB (Signed Release)

End of Technical Requirement Document · CaseFlow Engineering Group